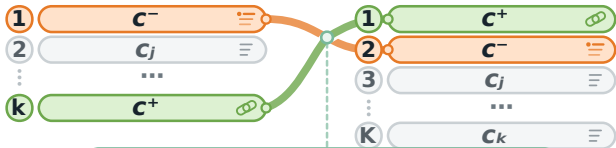


Initial retrieval for q_i
top-K $\Delta_i < 0$

Refined ranking
 $\Delta_i > 0$



Failure-Calibrated Refinement

$$\Delta_i = s(q_i, c^+) - s(q_i, c^-)$$

< 0



> 0



Paired code



Failure Candidate



Other candidate